

Homework 3, due Sep 10 before class

1. A sphere is charged with a surface charge $\sigma = \sigma_0 \cos \theta$. Find electric potential inside and outside of the sphere. Hint: since both inside and outside is vacuum, use expansion in spherical harmonics plus a jump at the surface.
2. Straight current wire has current distribution $j_z = j_0(1 - \varpi/\varpi_0)$ where $\varpi \leq \varpi_0$ is cylindrical coordinate. Find total current, vector potential and magnetic field.
3. Current density is given by

$$j_\phi = C_1 r^3 e^{-(2r/(3a))} \sin \theta \cos^2 \theta \quad (1)$$

where C_1 is some constant, spherical coordinates $r - \theta - \phi$. Find magnetic moment.